

Wall-Resolved Flux Pumping in the Class-A Adaptive Fermionic Circuit: Exact Flattened-Hamiltonian Benchmark and Implementation Handoff

Internal numerical methods and implementation note¹

¹*Class-A fermionic adaptive-circuit project*
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This document turns the proposed wall-charge pump into an implementation-ready experiment and records the exact flattened-Hamiltonian benchmark that motivates it. In the exact 20×40 reference, overlap-continued wall branches have signed spectral flow $(-1, +1)$ at $x = 5, 15$ and transfer one particle between wall basins to finite-size accuracy. Strict instantaneous ground-state occupation instead has zero net closed-cycle transfer. The apparent 0.98 versus 1 directional asymmetry is an endpoint convention at a 10^{-9} -scale physical-wall avoided crossing; a genuine reverse traversal undoes the pump to numerical precision. The earlier half- y entanglement-flow estimator fails its robustness gate and is not used.

For the monitored circuit, raw wall-charge change is not a conserved-current observable because feedback ancillas inject and remove particles. The required observable is a source-subtracted regional balance built from live charge and recorded gain/loss. We specify the flux schedule, canonical GPU API extension, observers, output schema, controls, tests, staged campaign, and claim ladder. The online Born-sampled pump is kept distinct from both frozen-record static replay and H1 modular-Hamiltonian propagation. This note is intended for direct handoff to an implementation agent.

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I. PURPOSE, AUTHORITY, AND STATUS

The class-A adaptive circuit of Bhuiyan, Pan, and Jian (BPJ) measures localized overcomplete-Wannier (OW) occupations and applies outcome-dependent fermionic feedback without postselection [1]. The question here is whether a controlled y holonomy produces a wall-resolved transverse pump. Table I fixes the claim boundary.

The production authority is `classA.U1FGTN.gpu.run_markov_circuit(...)`. An implementation must not reproduce its private cycle loop. The pre-existing `MICROSCOPIC_NUMERICAL_HALL_PROTOCOL.md` is a concise design note; this document adds the flattened- H audit and fixes the API and acceptance contract.

II. CONVENTIONS AND FORMALISM

A. Correlation matrix and trajectory

Following BPJ, the physical correlation matrix is

$$G_{ij} = \text{Tr}(\hat{\rho} \hat{\psi}_i^\dagger \hat{\psi}_j). \quad (1)$$

The GPU source calls its centered matrix $\mathbf{G}$; here that code-level object is

$$\Gamma = 2G - \mathbf{1}, \quad G = (\mathbf{1} + \Gamma)/2. \quad (2)$$

The symbol tr denotes a single-particle trace and Tr a many-body trace. A pure number-conserving Gaussian state has $G^2 = G$.

A record $\mathbf{m} = (m_1, \dots, m_D)$ produces the measurement evolution operator (mEO) $\hat{V}_{\mathbf{m}} = \hat{K}_{m_D} \cdots \hat{K}_{m_1}$. In the online protocol, every m_e is drawn from its instantaneous Born probability. No word is imposed.

B. Flux threading

Thread a y -cycle holonomy by

$$\hat{c}_{x,y+N_y,\mu} = e^{i\phi} \hat{c}_{x,y,\mu}. \quad (3)$$

The CPU exact construction uses

$$k_y(m, \phi) = 2\pi m/N_y + \phi/N_y, \quad (4)$$

TABLE I. Related constructions that must not be conflated.

Object	Meaning	Status
Exact flattened- H flow and charge	Quasistatic pure-state reference with continued wall branches; no monitored dynamics or Born records.	Implemented, audited
H3 half- y entanglement flow	Entanglement-spectrum estimator whose grid/tracking/cut gate failed.	Retired as evidence
Frozen-record twist replay	One imposed outcome word rerun at static twists; a conditional response, not an operational time-dependent pump.	Legacy pilot
H1 modular response	Propagation in modular time generated by an entanglement Hamiltonian; Separate probe neither monitored-circuit time nor charge pumping.	
Online monitored-channel pump	One uninterrupted Born-sampled trajectory with cycle-dependent controller twist and source subtraction.	Specified; unrun
Compiled electrical pump	Gate-resolved charge across a cut in a local system–ancilla dilation.	Future stronger construction

and closes by

$$H(\phi + 2\pi) = U_y H(\phi) U_y^\dagger, \quad [U_y]_{xy\mu, xy\mu} = e^{i2\pi y/N_y}. \quad (5)$$

GPU production uses its existing seam-gauge controller twist. A component acquires $e^{i\phi w}$ when its center-to-component path winds $w \in \{-1, 0, +1\}$ through the seam. Real-space diagonal regions and $\text{tr}(R_a P_e)$ are then flux independent. Reject a time-dependent uniform gauge until the associated temporal frame kick exists.

Increasing ϕ creates an azimuthal electromotive force; a Chern response is transverse and moves charge between the x -separated walls, as in Laughlin's Corbino argument [2, 3]. The real-space measurement does not require good k_y . Twisted boundaries remain the standard disorder/interacting formulation [4, 5].

C. Wall charge and chirality

Let $R_L + R_R = 1$. The exact calculation uses nearest-wall Voronoi basins at $x_L = 5, x_R = 15$, with ties split equally. The circuit uses

$$R_L : x = 0, \dots, 9, \quad R_R : x = 10, \dots, 19. \quad (6)$$

For slab $x \in [5, 15]$, the physical interface bonds are (4, 5) and (15, 16); the regional cuts lie in the bulk.

For a number-conserving exact state,

$$Q_a(\phi) = \text{tr}[R_a G(\phi)], \quad a = L, R, \quad (7)$$

$$q_{\text{pump}} = (\Delta Q_R - \Delta Q_L)/2. \quad (8)$$

Positive sign transfers charge from $x = 5$ to $x = 15$ under increasing twist. For a reference wall branch,

$$\partial_\phi E_w = N_y^{-1} \partial_{k_y} E_w = v_{y,w}/N_y. \quad (9)$$

The increasing-twist benchmark has

$$(\mathcal{F}_{x=5}, \mathcal{F}_{x=15}) = (-1, +1), \quad (10)$$

so $v_y < 0$ at $x = 5$ and $v_y > 0$ at $x = 15$; reversal flips both. Counterpropagating nonchiral pairs cancel in signed flow. This is direct chirality evidence, while charge transfer is its integrated real-space response [6, 7].

D. Purity

Regional charge is defined for pure or mixed G and is linear:

$$\text{tr}(R_a \overline{G}) = \overline{\text{tr}(R_a G)}. \quad (11)$$

Retain the trajectory distribution even when computing the mean from $\overline{G}$. Ordinary pumped charge is not generically integer for arbitrary mixed occupations. A purity gap can protect a mixed-state invariant without quantizing this linear charge [8, 9]; quantized open-system pumps require extra structure [10].

III. EXACT FLATTENED-HAMILTONIAN BENCHMARK

A. Operator, parameters, and provenance

For OW trial $\nu = A, B$ and band $n = \pm$, define the frame operator

$$F_{\nu n}(\phi) = \sum_{\mathbf{R}} |W_{\mathbf{R}, \nu n}(\phi)\rangle \langle W_{\mathbf{R}, \nu n}(\phi)|. \quad (12)$$

OW columns are nonorthogonal and overcomplete, so $F_{\nu n}$ is not generally an idempotent orthogonal projector. The one-body flattened operator is

$$H_{\text{flat}} = F_{A+} + F_{B+} - F_{A-} - F_{B-}. \quad (13)$$

All manifests hash `src/fgtn/classA_U1FGTN.py` to `2c8e86664d8d16da13b09670cc1cb78d2b0e14ff676b128d ff169cba1272f2c5`. Primary artifacts are

- `notebooks/flattened_hamiltonian_analysis/flattened_hamiltonian_twist_entanglement_benchmark.ipynb`;
- `notebooks/flattened_hamiltonian_analysis/exact_flattened_wall_charge_pump_refined.ipynb`;
- `notebooks/flattened_hamiltonian_analysis/outputs/exact_wall_charge_pump_refined_v1/run_20260901_002219/`;
- `notebooks/flattened_hamiltonian_analysis/run_wall_charge_pump_reversibility.py`;
- `notebooks/flattened_hamiltonian_analysis/outputs/exact_wall_charge_pump_reversibility_v1/run_20260901_162444/`.

B. Two state prescriptions

The instantaneous prescription reoccupies the globally lowest 800 eigenstates at every ϕ . It is the strict finite-size ground-state projector, closes under Eq. (5), and has zero *net endpoint* pump. Its intermediate regional charge is not constant.

The continued prescription chooses occupations once at ϕ_0 . In each k_y block it uses Hungarian one-to-one assignment to maximize

$$\sum_j |\langle u_j(\phi_i) | u_{\pi(j)}(\phi_{i+1}) \rangle|^2. \quad (14)$$

Within clusters $|E_i - E_j| < 10^{-9}$, an SVD polar rotation aligns the new and old subspaces. Occupations are never reselected. This is wall-localized diabatic continuation through the exponentially small wall anticrossing while remaining bulk-adiabatic. It is neither finite-time Schrödinger evolution nor the strict instantaneous ground state.

C. Validation and result

Errors are: Hermiticity 0; k_y off-block $< 9.5 \times 10^{-16}$; block reconstruction $< 7.1 \times 10^{-15}$; projector idempotency $< 2.3 \times 10^{-15}$; rank $< 2.3 \times 10^{-13}$; uniform/seam spectrum $< 7.4 \times 10^{-14}$; uniform/seam entanglement spectrum $< 1.3 \times 10^{-14}$; and Hamiltonian 2π closure $< 7.3 \times 10^{-15}$. Instantaneous projector closure is $< 1.2 \times 10^{-9}$ under its explicit 5×10^{-9} tolerance. Continued monodromy is $0.05 = \sqrt{2/800}$, one particle-hole replacement.

The continued wall branches have one signed flow per wall and no ambiguous crossings on 33/65/129 meshes. Because the strict finite spectrum anticrosses, Eq. (10) is the diabatic wall branch, not a literal instantaneous zero.

A window $|x - x_w| \leq 2$ misses only 9.3×10^{-7} (untruncated) or 1.3×10^{-6} (truncated) of the basin transfer. Localization fractions are 0.99999905 and 0.99999872. Thus the response is at the physical walls.

D. Endpoint artifact and reversal

The 10^{-7} start offset avoids a near-degenerate wall pair; it is not a Hamiltonian regulator. At $\phi = 0$, closest levels are

$$E = \pm 5.49020 \times 10^{-10} \text{ (untruncated)}, \quad E = \pm 7.51109 \times 10^{-10} \text{ (truncated)}. \quad (15)$$

TABLE II. Primary exact configuration.

Lattice/dimension	20×40 , $2N_x N_y = 1600$
Filling	fixed rank $N_x N_y = 800$
Mass/slab	$\alpha_{\text{top}} = 1$, $\alpha_{\text{triv}} = 30$, $x \in [5, 15]$
OW	$n_{\text{shell}} = 1$, trial X
Wall variants	dw.truncation=False, True
Twist meshes	both directions; 129, 257, endpoint-refined 751
Endpoint audit	$\phi_0 = \pm 10^{-7}$; $N_y = 16, 24, 40$

At $+10^{-7}$, the occupied near-zero state's left/right weights are (0.991985, 0.008014) and (0.990404, 0.009595), explaining 0.983972 and 0.980810.

Changing offset sign swaps which independent winding appears integer. At $N_y = 40$ without truncation,

$$\phi_0 = -10^{-7} : \quad q_+ = 0.9999999976, \quad q_- = -0.9839718413, \quad (16)$$

$$\phi_0 = +10^{-7} : \quad q_+ = 0.9839718590, \quad q_- = -0.9999999924. \quad (17)$$

The truncated case behaves identically with 0.98080969. Independent opposite windings from the same initial projector are not time reverses.

A true undo starts from the forward endpoint and traverses the cached path backward. Across $N_y = 16, 24, 40$, both offsets and truncations, $q_{\text{undo}} = -q_{\text{forward}}$ to stored precision; return error $\|G_{\text{return}} - G_0\|_F / (2N_x N_y)$ is $5.3 \times 10^{-22} - 4.2 \times 10^{-18}$. This validates tracking, not chirality. Future loops should put the crossing inside the path: use primary $\phi_\star = \pi$, with 0 as an origin control. That origin has not yet been run in the saved exact artifact.

E. Why H3 was retired

The half- y cut creates two entanglement endpoints on each of two walls. The audit varied 33/65/129/refined grids, 16/24/32/48 modes, five reference levels, wall/endpoint widths 1/2/3, seam, direction, state prescription, and full- y cuts. Physical wall flow passed, but entanglement flow did not. Saved classifications are grid/sensitivity plus tracking ambiguity (instantaneous) and grid/sensitivity plus cut mismatch (continued); all four cases have `genuine_flattened_state=False`. This rejects the estimator, not wall chirality. Direct wall charge replaces it.

IV. MONITORED-CHANNEL PUMP

A. Operational distinction

The primary experiment is one online Born-sampled ramp: equilibrate and then continue the same trajectory while changing twist once per complete Markov cycle. A fixed-record replay instead imposes one word in independent static reruns and is only a conditional response. H1 evolves in modular time and is an independent propagation test. These three outputs must never share a single pump label.

B. Source-subtracted balance

If event e measures $m_e \in \{0, 1\}$ and targets $s_e \in \{0, 1\}$, feedback transfers

$$\delta n_e = s_e - m_e. \quad (18)$$

Raw regional charge therefore mixes redistribution and injection. Define

$$N_a(c) = \text{tr}[R_a G(c)] = \frac{1}{2} \text{tr}[R_a (\mathbf{1} + \Gamma(c))], \quad (19)$$

$$A_a(c) = \sum_{c_0 < e \leq c} (s_e - m_e) \text{tr}(R_a P_e), \quad (20)$$

$$q_a(c) = N_a(c) - N_a(c_0) - A_a(c), \quad (21)$$

$$q_x(c) = [q_R(c) - q_L(c)]/2. \quad (22)$$

TABLE III. Endpoint wall charge at 20×40 , $\phi_0 = +10^{-7}$. Values are identical on 129/257/751 meshes: stability, not convergence of 0.98 to 1.

dw.truncation	$q_{\text{inst}}^{(+)}$	$q_{\text{cont}}^{(+)}$	$q_{\text{cont}}^{(-)}$	$ \Delta Q_L + \Delta Q_R _{\text{max}}$
False	1.492×10^{-8}	+0.9839718573	-0.9999999800	1.14×10^{-13}
True	2.523×10^{-9}	+0.9808096958	-1.0000000032	1.14×10^{-13}

Here $P_e = |\chi_e\rangle\langle\chi_e|$ is normalized and c_0 is the last equilibration cycle. Require

$$\Delta N_{\text{total}} = \sum_e (s_e - m_e), \quad q_L + q_R = 0 \quad (23)$$

to numerical tolerance.

This is a generalized monitored-channel pump, not literal electrical current: conditioned measurement innovation remains. A stronger electrical claim requires compiling every OW measurement into a fixed local Givens tree, performing the number measurement and filled/empty ancilla fermionic SWAP, uncomputing the tree, and counting charge across a spatial cut gate by gate. Validate two trees and two cut placements.

C. Deterministic mean screen

For normalized P , $Q = \mathbf{1} - P$, and target s , perfect correction gives

$$\overline{G}' = Q\overline{G}Q + sP, \quad \overline{\delta n} = s - \text{tr}(P\overline{G}). \quad (24)$$

`mean_replacement=True` implements the first but does not yet emit the second. Add that source observer and run the deterministic screen before physical trajectories. It is not a substitute for their distribution.

V. RAMP AND CAMPAIGN ARMS

Let $T_{\text{eq}} = 2N_y$ and T_ϕ be an *additional* pump duration. For orientation $\sigma = \pm 1$,

$$f(u) = u - \frac{\sin(2\pi u)}{2\pi}, \quad f'(0) = f'(1) = 0, \quad (25)$$

and define an absolute schedule of length $T_{\text{eq}} + T_\phi + 1$:

$$\phi_c = \begin{cases} \phi_\star, & 0 \leq c \leq T_{\text{eq}}, \\ \phi_\star + \sigma 2\pi f[(c - T_{\text{eq}})/T_\phi], & T_{\text{eq}} < c \leq T_{\text{eq}} + T_\phi. \end{cases} \quad (26)$$

Use primary $\phi_\star = \pi$ and repeat the pilot at 0. A hold arm remains at $\phi_\star$. Apply ϕ_c before the first update of cycle c ; baseline at $c_0 = T_{\text{eq}}$ without reinitialization. Use fresh random schedules and instantaneous Born outcomes.

Fix $n_{\text{shell}} = 1$, $\alpha_{\text{top}} = 1$, $\alpha_{\text{triv}} = 30$, trial X , pure half filling, perfect correction, complex128, random order, and no postselection. Compare:

1. soft explicit interface: `dw_truncation=False`, `meas_slab_only=False`;
2. hard support termination: both flags `True`;
3. matched trivial: $\alpha_{\text{top}} = \alpha_{\text{triv}} = 30$;
4. $+2\pi$, -2π , and hold.

Record `DW_loc`. GPU default gives $[5, 15]$ for $N_x = 20$, CPU default gives $[4, 16]$; CPU parity must pass `dw_interval=(5, 15)`.

Use

$$C_{\text{odd}}(T_\phi) = \frac{\varsigma}{2} [\overline{q_x(+2\pi)} - \overline{q_x(-2\pi)}], \quad (27)$$

with geometry sign ς fixed from the exact reference. Retain even and hold responses. Fit $C_{\text{odd}} = C_\infty + a/T_\phi$ and test b/T_ϕ^2 rather than assuming the asymptote.

VI. CANONICAL IMPLEMENTATION CONTRACT

A. GPU API

The GPU class already has the absolute setter

```
set_controller_twist(phi, gauge="seam", seam_y=None)
```

but no schedule. Add to `src/fgtn/classA_U1FGTN_gpu.py`:

```
controller_twist_schedule=None
controller_twist_gauge="seam"
controller_twist_seam_y=None
mean_transfer_observer=None
```

with this contract:

1. Schedule is `None` or a finite 1D array of absolute unwrapped fluxes of length `cycles + 1`; reject all else.
2. Reset to `schedule[0]` before cycle-zero observation in every batch. Calls are absolute, not cumulative.
3. Apply `schedule[c]` before any site update in cycle c and hold it for that complete cycle.
4. Permit dynamic schedules only in seam gauge until the uniform-gauge temporal kick exists.
5. Store schedule or cryptographic hash, gauge, seam, protocol version, and `DW_loc` in cache identity, configuration, checkpoints, and manifest. Static twist must also enter cache identity.
6. Resume only on an exact schedule-hash match; preserve old behavior bit-for-bit when the schedule is `None`.

A callback can set the next cycle in a one-batch pilot, but production must not use it: batching, resume, random streams, and metadata are fragile. Do not call the engine one cycle at a time. The setter currently rebuilds the full local cache at every call. Benchmark this and optimize only with fresh-rebuild parity.

B. CPU/GPU distinction and observers

CPU `construct_OW_projectors(..., twist_y=phi)` reconstructs from the shifted momentum grid. GPU seam-dresses an existing finite-support controller. These are distinct finite-support protocols. Use GPU seam twist for production and CPU uniform reconstruction for exact references and explicit tests.

For pure occupied-frame trajectories, compute N_L, N_R from squared live frame-row norms in `native_cycle_observer`; do not materialize dense covariances each cycle. For mean/covariance runs use Eq. (2).

Existing `record_observer` emits `transfer = target_occupied - outcome_occupied`. Precompute $w_{a,e} = \text{tr}(R_a P_e)$ by site/channel in seam gauge. Add `mean_transfer_observer` emitting

```
cycle, update_index, site_id, channel_label, sample_index,
target_occupation, occupation_probability,
expected_transfer = target_occupation - occupation_probability
```

per event. Public `mean_replacement_channels(...)` supplies ordered targets and spinors for regional weights.

C. Reference ordering

```
schedule = make_schedule(phi_star, sign, T_eq, T_phi) # cycles+1
regions = make_regions(DW_loc)
weights = precompute_region_weights(model, regions, gauge="seam")
model.run_markov_circuit(
    cycles=T_eq + T_phi,
    controller_twist_schedule=schedule,
    controller_twist_gauge="seam",
    G_history=False, save=False,
```

```

state_representation="physical_frame",
native_cycle_observer=regional_charge_observer,
record_observer=regional_source_observer,
perfect_correction=True, postselect=False,
sequence="random",
meas_slab_only=case.meas_slab_only,
random_seed=seed,
)
# Baseline at c0=T_eq; compute q_L, q_R, q_x, continuity.

```

For `mean_replacement=True`, use covariance representation and `mean_transfer_observer`; keep definitions identical.

D. Outputs and integration

Each versioned run must save:

1. manifest: source hash, canonical entry point, protocol, device/dtype, geometry, flags, filling, initialization, correction, seeds, schedule/hash, gauge, seam, origin;
2. cycle table: sample, cycle, ϕ , N_L , N_R , A_L , A_R , q_L , q_R , q_x , total charge/rank, continuity residuals;
3. endpoint table: +, −, hold, odd/even response, rate, case, flags;
4. compact event-source audit keyed by cycle/site/channel; and
5. PDF and 300-dpi PNG plots of $q_x(c)$, raw charge versus source, rate scaling, arms, final x density, and trajectory distribution.

Do not save dense histories or discard trajectory-level endpoints.

All simulation behavior belongs in canonical GPU source. Synchronize every `classA_U1FGTN_gpu.py` outside `erroneous_gpu_stuff/`. A new Colab bundle belongs in `00_WORKSPACE/CURRENT/final_production_new_designs/`, must be registered, import the engine, show progress, and use the standard disconnect cell.

VII. TESTS AND ACCEPTANCE GATES

Automate before physics runs:

1. schedule shape/finite validation and exact cycle-timing spy;
2. per-batch reset and absolute/idempotent setter;
3. constant-schedule/static-setter parity and $\phi = 0$ parity;
4. seam 2π closure, seam relocation, unchanged norms/supports/weights;
5. local-cache update versus fresh rebuild;
6. local/dense and covariance/physical-frame parity;
7. rejection of dynamic uniform gauge;
8. per-trajectory Eq. (23);
9. mean replacement versus explicit Born enumeration;
10. schedule hash in cache/save/resume; and
11. explicit geometry parity without claiming CPU/GPU twist identity.

Do not claim a plateau unless every trajectory passes source continuity, +/− odd response is stable, hold/even are small or modeled, matched trivial is null, signs reverse, results survive rate/seam/region/origin controls, density change is wall localized, soft and hard walls are reported separately, and means include trajectory distributions and confidence intervals. Do not impose an integer before rate, size, gauge, origin, region, wall-protocol, and purity systematics.

VIII. STAGED CAMPAIGN

If the mean screen has no odd source-subtracted response, diagnose the observable, seam, and controller before trajectories. If records are broad, report the distribution. Never reweight ordinary Born samples by stored likelihood; that double-counts Born weight.

IX. CLAIM LADDER

1. *Exact reference*: opposite signed flattened- H wall flow and nearly one fixed-label transfer, with endpoint convention stated.
2. *Mean monitored channel*: nonzero orientation-odd source-subtracted balance under the exact affine mean map.
3. *Physical trajectories*: stable ordinary-Born mean and quantified distribution under the online ramp.
4. *Chirality synthesis*: orientation agreement among pump, H1/H2 propagation, and exact wall flow; no circuit k_y is required.
5. *Electrical Hall response*: only after local gate compilation.
6. *Quantization*: only after rate, size, gauge, origin, region, purity, and concentration studies.

The exact reference alone is not monitored-trajectory evidence, and true undo alone proves only numerical reversibility. Retired H3 is not corroboration.

X. RELATED WORK

The closest precedent is Julià-Farré *et al.*, who thread flux through an inhomogeneous Chern cylinder and infer domain Chern numbers from left/center/right charge [11]. Their DMRG stripe differs from this circuit, but the regional accounting is directly analogous.

Foundations are Laughlin, TKNN, Thouless, and twisted-boundary disorder extensions [2, 4, 5, 12, 13]. Hatsugai and Halperin connect invariants to edge flow [3, 6]. Mixed/open caveats are in Refs. [8–10, 14]. Experiments include ultracold fermion/boson pumps [15, 16], disorder-dependent pumping [17], an atomic Hall cylinder [18], and a quantum anomalous Hall Corbino pump [19]; see also the review [20]. To our knowledge, prior work does not combine adaptive Born trajectories, wall-resolved source subtraction, and a flattened- H comparison. Any novelty statement must retain that qualifier.

XI. IMPLEMENTATION-AGENT COMPLETION CHECKLIST

The next agent must return:

1. canonical GPU schedule and mean-source APIs with docstrings and no duplicated cycle loop;
2. all Sec. VII tests passing;
3. synchronized active GPU copies and synchronization count;

TABLE IV. Recommended staging. $T_{\text{eq}} = 2N_y$ is additional to T_ϕ . Production shards contain five independent trajectories.

Stage	Geometry	T_ϕ	Required arms
Smoke	12×16 , one	N_y	Soft interface, $+/-$ /hold; timing, continuity, static parity.
Mean screen	20×24 , five schedules	$4N_y, 8N_y, 16N_y$	Soft/hard, $+/-$ /hold, trivial; both origins at pilot rate.
Trajectory pilot	20×24 , ten	$4N_y, 8N_y, 16N_y$	Forward all rates; reverse, hold, trivial at $8N_y$.
Production	20×24 , 25	$4N_y, 8N_y, 16N_y$	Same preregistered arms, shards of five.
Size check	20×32 , 25	$8N_y$ first	Soft/hard plus matched trivial, $+/-$ /hold.

4. a registered, versioned Colab bundle;
5. a local smoke artifact with manifest, cycle/endpoint tables, source audit, and figures;
6. explicit timing, batch-reset, continuity, static-parity, closure, and hold/trivial-null evidence; and
7. a statement of what remains unrun. Passing software tests is not a completed physics campaign.

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